# PREVENTATIVE MAINTENANCE CHECKLIST

| Enclosed Cab | Underground Haulage Truck | Tier 4 | Dump |
|---|---|---|---|

| 2000 | Hour |
|---|---|

DATE: _____

UNIT NUMBER: _____

TECHNICIANS: _____

HOURS: _____

WORK ORDER: _____

***** PM TO BE COMPLETED ONCE THE UNIT IS WASHED *****

***Note: All specifications, procedures, and parts not identified in this PM can be found in the OEM manuals***

## PRE-WORK SAFETY TASKS

### MANDATORY PPE

**LOCKOUT / TAGOUT** – Ensure all procedures are followed to eliminate stored energy

Before working on brake system or hydraulic system drain accumulator pressure

Ensure deck, walkways and steps are free from foreign material such as oil and debris and that traction surface is in place and in good condition

Read and understand all warning plates on machine before operating, lubricating or repairing machine

To avoid burns, be alert for hot parts on machines, which have just been stopped and have hot fluids in lines, tubes and compartments.

Battery electrolyte contains acid – avoid contact with skin or eyes.

Perform all running checks in an open area free of obstruction and personnel

Ensure the box is locked in the raised position and the steering lock is in place before working on the machine

Ensure the box is free of muck before working on the machine

If working on air system brakes, bleed down air system

Ensure moldboard and ripper are lowered to the ground before servicing equipment

Refer to manufacturer's operation and maintenance manuals for equipment specifications, fluid capacities, operating procedures and maintenance procedures

Be sure the area around the machine is clear of personnel and obstructions while performing running checks

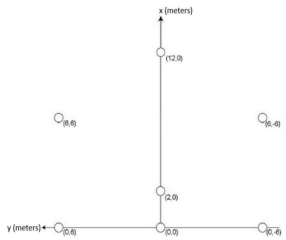
***FOR MORE COMPLETE LIST OF SAFETY MESSAGES, REFER TO THE OPERATION AND MAINTENANCE MANUAL FOR THIS MACHINE***

| A = OK, B = DEFECT FOUND AND REPAIRED, C = DEFECT FOUND NOT REPAIRED, D = N/A<br>⚠ = TAKES MACHINE OUT OF SERVICE  💧 = CONDITION MONITORING POINT | | A | B | C | D | ⚠ 💧 |
|---|---|---|---|---|---|---|
| **PRE-ISOLATION TASKS AND INITIAL WALK AROUND** | | | | | | |
| 1 | Wash unit (remove belly pan before washing engine area) | | | | | |
| 2 | Pre-op book - check unit has book & ensure book is current & correct | | | | | |
| 3 | Pre-op book comments - review | | | | | |
| 4 | Check condition of guard rails, ensure they all unfold and lock in place | | | | | |
| 5 | Check "STORED ENERGY" labels, make sure they are in place and legible | | | | | |
| 6 | Ensure the unit is equipped with two wheel chocks, that they are in good condition, and that the storage mounts on the machine are in good shape | | | | | |
| 7 | Check condition & operation access doors, lids & prop rods, verify that all traction surfaces are in good shape and the steps and grab handles are safety green | | | | | |
| 8 | Check front & rear frames for cracks & report damage | | | | | ⚠ |
| 9 | Check the frame rub bars for excess wear | | | | | |
| **OPERATOR STATION** | | | | | | |
| 10 | Check proper operation of the dash alarm buzzer, vehicle horn and backup alarm | | | | | ⚠ |
| 11 | Test the operation of the neutral lockout (try to start machine while in gear) | | | | | ⚠ |
| 12 | Test the operation of the function lockout | | | | | ⚠ |
| 13 | Gauges, dash lights, marker lights, running lights - check condition & operation | | | | | |
| 14 | Operator control labels - ensure they are in place & legible | | | | | |
| 15 | Check proper operation of the cab door gas spring assembly and ensure the rubber stops are in good condition | | | | | |
| 16 | Labels - inspect labelling on unit (CFM, warning, operator labels) | | | | | |
| 17 | Brake test procedure plaque - check if it is in place | | | | | |
| 18 | Brake pedal - check for smooth operation, wear of pivot & remove any obstructions on floor | | | | | ⚠ |
| 19 | Ensure ABA is functioning properly | | | | | ⚠ |
| 20 | Service brake holding capacity - test | | | | | ⚠ |
| 21 | Park brake holding capacity - test | | | | | ⚠ |
| 22 | Check service brake pressure | | | | | ⚠ |
| 23 | Throttle - check for smooth operation & remove any obstructions on floor | | | | | ⚠ |
| 24 | Decelerator - check for smooth operation & remove any obstructions on floor | | | | | ⚠ |
| 25 | Perform ECM download, provide download file to Planners / Reliability | | | | | |

| A = OK, B = DEFECT FOUND AND REPAIRED, C = DEFECT FOUND NOT REPAIRED, D = N/A<br>⚠ = TAKES MACHINE OUT OF SERVICE  ♦ = CONDITION MONITORING POINT | | A | B | C | D | ⚠ ♦ |
|---|---|---|---|---|---|---|
| 26 | Seat and armrests – check overall condition and ensure the height and suspension adjusters are working | | | | | |
| 27 | Seatbelt – check condition & expiry date | | | | | ⚠ |
| 28 | ROPS / FOPS – check for damage & loose mounting bolts | | | | | |
| 29 | Hydraulic & transmission control levers – check for free movement & looseness | | | | | ⚠ |
| 30 | Check condition of the door, hinges, latches, limiting strap, and door seals | | | | | |
| 31 | Oil leaks in cab & under door panel – check for leaks | | | | | |
| 32 | Check condition of windsheild wipers and verify the wipers and washers are working | | | | | |
| 33 | Windows – check condition and ensure the emergency egress tab is in place | | | | | ⚠ |
| 34 | Check operation of the heating / air conditioning system including the fan speed and temperature switches, ensure the evaporator drains are clear and the directional vents are present and in good condition | | | | | |

## RUNNING CHECKS

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 35 | Engine breather tube – check for excessive smoke or leaking engine oil | | | | | | |
| 36 | Transmission – check for oil leaks | | | | | | |
| 37 | Forward / reverse – operate unit, listen for noise and ensure that all gears are functioning | | | | | | ⚠ |
| 38 | Perform torque converter idle and stall test | | | | | | ⚠ |
| 39 | Verify operation of the torque lockup | | | | | | |
| 40 | Verify all hydraulic functions are working, including float | | | | | | |
| 41 | Check operation of the payload monitoring system, calibrate system | | | | | | |
| 42 | Ensure the box up gear lockout is working correctly | | | | | | |
| 43 | Verify operation of the brake retarder / compression brake | | | | | | ⚠ |
| 44 | Check for play in the articulation pins, ensure that all of the bolts are tight | | | | | | ⚠ |
| 45 | Steering stop – check clearance both ways to avoid steering cylinders bottoming out, adjust steering stops and neutralizer valves if necessary | | | | | | |
| 46 | Check steering cycle times at high idle (sec) | Left to Right | | Right to Left | | | ♦ |
| 47 | Verify secondary (emergency) steering is operational | | | | | | |
| 48 | Check for play (including side to side play) in the steering cylinder pins and bushings. Ensure pin retainers are tight and in place | | | | | | |
| 49 | Check for play (including side to side play) in the lift cylinder pins and bushings. Ensure pin retainers are tight and in place | | | | | | |

| A = OK, B = DEFECT FOUND AND REPAIRED, C = DEFECT FOUND NOT REPAIRED, D = N/A<br>⚠ = TAKES MACHINE OUT OF SERVICE  ♦ = CONDITION MONITORING POINT | | | | | A | B | C | D | ⚠ ♦ |
|---|---|---|---|---|---|---|---|---|---|
| 50 | Check lift cylinder cycle time (sec) | Seated to fully raised | | Fully raised to seated | | | | | ♦ |
| 51 | Check lift cylinder drift rate | | | | | | | | |
| 52 | Check the operation of the backup camera, ensure the camera is aimed properly<br>Position pylon 3 under camera, pylon 2 at 6' (2m) directly in line with 3, and pylon 1 direcly in line with 2 and 3 as far away as possible<br>Position pylons 4 and 5 at 6' (2m)to the right and left of the camera centerline respectively and halfway between pylons 1 and 2<br>Position pylons 6 and 7 directly 6' (2m) in line left and right respectively with the camera<br>Note: Position as shown in illustration. Camera view (illustration 2)<br>Pylons 1, 2, and 3 must be in a straight line as per straight line in camera view, Pylons 4, and 5 must be the same distance to the outside of the monitor as per arrows in camera view, Pylons 6 and 7 must not be seen by the camera<br>  <br> | | | | | | | | |
| 53 | Perform forced regeneration | | | | | | | | |
| 54 | Check brake accumulator charge valve settings | | | | | | | | ⚠ |
| 55 | Perform pilot oil accumulator leak down test | | | | | | | | ⚠ |
| 56 | Perform pilot oil accumulator pressure test | | | | | | | | ⚠ |
| 57 | Check steering relief pressure | | | | | | | | ⚠ |
| 58 | Check dump relief pressure | | | | | | | | ⚠ |
| 59 | Test park brake tow release system | | | | | | | | ⚠ |

## ISOLATION

| | | | | | | |
|---|---|---|---|---|---|---|
| 60 | Lift machine and place on jack stands (or ramps) | | | | | |
| 61 | Bleed hydraulic and brake pressures | | | | | |
| 62 | Install steering lock and boom lock | | | | | |
| 63 | Tag and lockout machine | | | | | |
| 64 | Barricade working area | | | | | |
| 65 | Ensure cabin tilt function is working properly and that the cylinder is in good condition | | | | | |
| 66 | Install cabin tilt lock | | | | | |

| A = OK, B = DEFECT FOUND AND REPAIRED, C = DEFECT FOUND NOT REPAIRED, D = N/A<br>⚠ = TAKES MACHINE OUT OF SERVICE 💧 = CONDITION MONITORING POINT | | A | B | C | D | ⚠ 💧 |
|---|---|---|---|---|---|---|
| **OIL SAMPLES + OIL AND FILTER CHANGES** | | | | | | |
| 67 | Take engine oil sample, change oil, inspect magnetic plug<br>Rate magnetic plug from 1 – 5 based on "Magnetic Plug Ratings" | | | | | 💧 |
| 68 | Replace engine oil filter, cut old filter and inspect<br>Rate filter debris from 1 – 4 based on "Filter Contamination Scale" | | | | | 💧 |
| 69 | Replace water seperator & fuel filter | | | | | |
| 70 | Replace engine fuel filter | | | | | |
| 71 | Engine coolant – take sample, check coolant level | | | | | |
| 72 | Replace engine coolant filter | | | | | |
| 73 | Change transmission filter, cut old filter and inspect<br>Rate filter debris from 1 – 4 based on "Filter Contamination Scale" | | | | | 💧 |
| 74 | Clean transmission magnetic screen | | | | | |
| 75 | Take sample and change transmission oil, inspect magnetic plug<br>Rate magnetic plug from 1 – 5 based on "Magnetic Plug Ratings" | | | | | 💧 |
| 76 | Change gearbox filter, cut old filter and inspect<br>Rate filter debris from 1 – 4 based on "Filter Contamination Scale" | | | | | 💧 |
| 77 | Take sample and change gearbox oil, inspect magnetic plug<br>Rate magnetic plug from 1 – 5 based on "Magnetic Plug Ratings" | | | | | 💧 |
| 78 | Replace (crankcase) fumes disposal filter (OCV) | | | | | |
| 79 | Clean DEF filler screen | | | | | |
| 80 | Replace DEF filter | | | | | |
| 81 | Change hydraulic (return) filter, cut old filter and inspect<br>Rate filter debris from 1 – 4 based on "Filter Contamination Scale" | | | | | 💧 |
| 82 | Change brake filter, cut old filter and inspect<br>Rate filter debris from 1 – 4 based on "Filter Contamination Scale" | | | | | 💧 |
| 83 | Take sample and change hydraulic oil, inspect magnetic plug<br>Rate magnetic plug from 1 – 5 based on "Magnetic Plug Ratings" | | | | | 💧 |
| 84 | Front and rear differential oil – take sample, change oil, check magnetic plugs<br>Rate magnetic plug from 1 – 5 based on "Magnetic Plug Ratings" | | | | | 💧 |
| 85 | Take oil sample, change final drive oil, inspect magnetic plugs<br>Rate magnetic plug from 1 – 5 based on "Magnetic Plug Ratings" | | | | | 💧 |
| 86 | Take samples and change final drive oil (all four corners), inspect magnetic plugs | | | | | |

| A = OK, B = DEFECT FOUND AND REPAIRED, C = DEFECT FOUND NOT REPAIRED, D = N/A<br>⚠ = TAKES MACHINE OUT OF SERVICE  💧 = CONDITION MONITORING POINT | | A | B | C | D | ⚠ 💧 |
|---|---|---|---|---|---|---|
| 87 | Ensure proper labelling and packaging of samples to be sent out for analysis | | | | | |
| 88 | Replace primary & secondary air filters | | | | | |
| 89 | Replace cab air filter | | | | | |
| 90 | Replace recirculating air (A/C) filter | | | | | |

## ENGINE COMPARTMENT

| | | | | | | |
|---|---|---|---|---|---|---|
| 91 | RADIATOR – check condition & tightness of mount bolts, inspect fins for damage and verify the fins are not plugged and the cores are not leaking | | | | | |
| 92 | AFTERCOOLER – check condition & tightness of mount bolts, inspect fins for damage and verify the fins are not plugged and the cores are not leaking | | | | | |
| 93 | HYDRAULIC COOLER – check condition & tightness of mount bolts, inspect fins for damage and verify the fins are not plugged and the cores are not leaking | | | | | |
| 94 | AXLE OIL COOLER – check condition & tightness of mount bolts, inspect fins for damage and verify the fins are not plugged and the cores are not leaking | | | | | |
| 95 | AIR CONDITIONING CONDENSOR – check condition & tightness of mount bolts, inspect fins for damage and verify the fins are not plugged and the cores are not leaking, and ensure fans are secured to the condensor body and working | | | | | |
| 96 | Adjust valve clearance, record before measurements<br>Adjust clearance when deviation exceeds +/-0.002" | | | | | 💧 |
| | IN 1A IN 1B IN 2A IN 2B IN 3A IN 3B IN 4A IN 4B IN 5A IN 5B IN 6A IN 6B | | | | | 💧 |
| | | | | | | 💧 |
| | 1 2 3 4 5 6 | | | | | 💧 |
| | | | | | | 💧 |
| | EX 1A EX 1B EX 2A EX 2B EX 3A EX 3B EX 4A EX 4B EX 5A EX 5B EX 6A EX 6B | | | | | 💧 |
| 97 | Check injector height adjustment | | | | | |
| 98 | Adjust compresson brake | | | | | |
| 99 | INTAKE – check condition of overall system, including filter housing and precleaner, piping to and from engine and aftercooler, and the intake manifold (check all components for loose clamps, bolts, nuts, etc and rubber hoses for degredation, and all components | | | | | ⚠ |
| 100 | EXHAUST – check condition of overall system including piping, clamps, purifiers and exhaust wrap on piping | | | | | |

| A = OK, B = DEFECT FOUND AND REPAIRED, C = DEFECT FOUND NOT REPAIRED, D = N/A<br>⚠ = TAKES MACHINE OUT OF SERVICE 💧 = CONDITION MONITORING POINT | | | A | B | C | D | ⚠ 💧 |
|---|---|---|---|---|---|---|---|
| 101 | Remove the intake and exhaust connections and check condition of turbocharger (bearings, impellers, and housings), and ensure intake is free of oil | | | | | | ⚠ |
| 102 | Check torque on impeller and turbine housing clamps (loose clamps may be a quick indicator of internal issues) and check for any evidence of oil coming from the intake and exhaust housings | | | | | | ⚠ |
| 103 | Replace DPF, send old one out to be cleaned<br>Record PO that the DPF is sent out on for cleaning | PO | | | | | 💧 |
| 104 | Replace NRS cooler, send old one out to be cleaned and pressure tested | | | | | | |
| 105 | Replace EGR valve, send old one out to be cleaned | | | | | | |
| 106 | BATTERIES - check all terminals and posts are free from corrosion and securely fastened | | | | | | |
| 107 | Check level of electrolyte in the battery and test specific gravity (if battery is not maintenance free) | | | | | | |
| 108 | Check for loose or corroded wiring on battery and master disconnect switch | | | | | | |
| 109 | Check condition of all wiring loom condition & routing | | | | | | |
| 110 | Check starter mounting bolts and wiring connections | | | | | | |
| 111 | Perform starter draw test | | | | | | |
| 112 | Check for loose or corroded wiring on starter | | | | | | |
| 113 | Check alternator mounting bolts and wiring | | | | | | |
| 114 | Check for loose or corroded wiring for alternator | | | | | | |
| 115 | Check belt tension and condition | | | | | | ⚠ |
| 116 | Replace engine belts | | | | | | |
| 116 | Check condition of pulleys, fan and shrouds | | | | | | |
| 118 | Check condition of all hoses for leaks | | | | | | |
| 119 | Check condition of engine mounts & tightness of bolts | | | | | | ⚠ |
| 120 | Inspect fuel lines for leaks and clearance, inspect fitment of all clamps | | | | | | |
| 121 | Check for leaks from water pump, hoses, radiator and overall cleanliness of cooling system | | | | | | |
| 122 | Check condition of engine covers and check all guards are in place & secure | | | | | | |
| 123 | Oil & fuel leaks - check engine bay for leaks & cleanliness | | | | | | |
| 124 | Hydraulic, transmission and brake/steering pumps - check for leaks | | | | | | |

## TIRES

| A = OK, B = DEFECT FOUND AND REPAIRED, C = DEFECT FOUND NOT REPAIRED, D = N/A<br>⚠ = TAKES MACHINE OUT OF SERVICE  ♦ = CONDITION MONITORING POINT | | A | B | C | D | ⚠ ♦ |
|---|---|---|---|---|---|---|
| 125 | Check tires for large sections missing from the face, face peeling, large cuts (cords exposed), sidewall damage (impact damage), sidewall cuts (cords exposed), or any tire bulging | | | | | ⚠ |
| 126 | Check condition of rims for cracks, bends, missing or loose studs / nuts, elongated bolt holes, damage to tire lock ring | | | | | ⚠ |
| 127 | Please record serial numbers from the tires<br>2D ____ 3D ____<br>1B ____ 2C ____ 3C ____<br>1A ____ 2B ____ 3B ____<br>2A ____ 3A ____ | | | | | |
| 128 | Measure inside & outside tire depth, record lowest reading (in /32")<br>2D ____ 3D ____<br>1B ____ 2C ____ 3C ____<br>1A ____ 2B ____ 3B ____<br>2A ____ 3A ____ | | | | | ♦<br>♦<br>♦<br>♦<br>♦<br>♦ |
| 129 | Measure diameter with tape measure across center of tire (ensure there is no more than 2% difference side to side and 4% difference front to back)<br>2D ____ 3D ____<br>1B ____ 2C ____ 3C ____<br>1A ____ 2B ____ 3B ____<br>2A ____ 3A ____ | | | | | |
| 130 | Check inflation pressure in tires<br>2D ____ 3D ____<br>1B ____ 2C ____ 3C ____<br>1A ____ 2B ____ 3B ____<br>2A ____ 3A ____ | | | | | ♦<br>♦<br>♦<br>♦<br>♦<br>♦ |

## POWERTRAIN

| | | | | | | |
|---|---|---|---|---|---|---|
| 131 | Check condition of transmission mounts & torque monting bolts | | | | | ⚠ |
| 132 | Check for play in the oscillation bushing, ensure that all of the bolts are tight | | | | | ⚠ |

| A = OK, B = DEFECT FOUND AND REPAIRED, C = DEFECT FOUND NOT REPAIRED, D = N/A<br>⚠ = TAKES MACHINE OUT OF SERVICE  ♦ = CONDITION MONITORING POINT | | A | B | C | D | ⚠ ♦ |
|---|---|---|---|---|---|---|
| 133 | Inspect all universal joints for wear and play, check & tighten bolts | | | | | ⚠ |
| 134 | Check tighteness of oscillating axle bolts, check for play in the oscillation bushings | | | | | |
| 135 | Check front, centre, and rear axle mounting frame pivot pins and bushings for play, ensure they are taking grease | | | | | ⚠ |
| 136 | Check front centre, and rear, stabilizer links for excess play or damage | | | | | ⚠ |
| 137 | Check suspension cylinders for leaks or damage, measure suspension height to ensure the cylinder is still in good condition | | | | | ⚠ |
| 138 | Check front and rear axle mounting bolts using hammer (tapping) to identify any loose bolts | | | | | ⚠ |
| 139 | Check the nitrogen pressure in the brake accumulators | | | | | |
| 140 | | | | | | |
| 141 | | | | | | |
| 142 | | | | | | |
| 143 | Check torque / transmission compartment for any leaks, check hoses for wear, damage and proper routing, check electrical harnesses for proper routing and any abrasion | | | | | |
| 144 | Check condition of ground strap in the articulation hitch | | | | | |

## DUMP BOX / EJECTOR BOX

| | | | | | | |
|---|---|---|---|---|---|---|
| 145 | Check box for wear and cracks | | | | | |
| 146 | Check for uneven wear or gaps on box rest support pads | | | | | |
| 147 | Ensure grease hoses are in good condition and all pins are taking grease | | | | | ⚠ |
| 148 | Inspect condition dump cylinders | | | | | |
| 149 | Take and record 5 measurements 6" from the position indicated<br>*Replace liners when curved pieces reach 6mm or .250"* | | | | | ⚠ |
| | Pos 1 Left Front Fender Top Actual | | | | | ♦ |
| | Pos 2 Front Floor Actual | | | | | ♦ |
| | Pos 3 Right Front Fender Top Actual | | | | | ♦ |
| | Pos 4 Left Centre Fender Top Actual | | | | | ♦ |
| | Pos 5 Centre Floor Actual | | | | | ♦ |
| | Pos 6 Right Centre Fender Top Actual | | | | | ♦ |
| | Pos 7 Left Rear Fender Top Actual | | | | | ♦ |
| | Pos 8 Rear Floor Actual | | | | | ♦ |
| | Pos 9 Right Rear Fender Top Actual | | | | | ♦ |

## HYDRAULICS AND HYDRAULIC TANK COMPARTMENT

| A = OK, B = DEFECT FOUND AND REPAIRED, C = DEFECT FOUND NOT REPAIRED, D = N/A<br>⚠ = TAKES MACHINE OUT OF SERVICE 💧 = CONDITION MONITORING POINT | | A | B | C | D | ⚠ 💧 |
|---|---|---|---|---|---|---|
| 150 | Check condition & routing of all hoses in hydraulic tank bay | | | | | |
| 151 | Check condition & routing of all hoses in articulation area | | | | | |
| 152 | Check condition & routing of all hoses in rear frame area | | | | | |
| 153 | Check condition & installation of hydraulic screen | | | | | |
| 154 | Check hydraulic tank condition & mounting, check for leaks | | | | | |
| 155 | Check all valves for leaks | | | | | |
| **GREASING** | | | | | | |
| 156 | Check level of the autogreaser reservoir | | | | | |
| 157 | Check for proper operation of grease pump | | | | | ⚠ |
| 158 | Check for leaks at all lines & connections on auto lube system, if equipped | | | | | ⚠ |
| 159 | Grease fan bearing on engine | | | | | |
| 160 | Check for proper grease distribution at all auto lube points | | | | | |
| 161 | Grease all universal joints, slip joints & pillow block | | | | | |
| 162 | Grease all cover and door hinges | | | | | |
| **POST WORK TASKS** | | | | | | |
| 163 | Recheck all fluid levels and verify all drain and level plugs are tight | | | | | |
| 164 | Make sure every closure, access panel, cover, and door is properly closed and locked | | | | | |
| 165 | Clean work area of debris, old parts, excess / unused materials and lubricants, roll up hoses and cords, return chemicals to flammable cabinet, etc | | | | | |
| 166 | Return equipment worked on or inspected to a safe operating condition | | | | | |
| 167 | Ensure work orders are submitted to planning for all found corrective issues to be resolved at a later date | | | | | |
| 168 | Ensure all reports and paperwork is signed off, dated and returned to Supervisor / Planning | | | | | |
| 169 | Walk around equipment to ensure that all foreign materials are cleaned up & put away properly. (rags, tools, blocks, chains, parts, garbage) | | | | | |
| 170 | Complete paperwork – ensure all work orders are filled in properly, found work scope sheets are filled out & handed into supervisor for review | | | | | |
| 171 | Complete parts & core returns | | | | | |
| 172 | Clean up old filters / oils and dispose of properly | | | | | |
| **DE-ISOLATION** | | | | | | |

| A = OK, B = DEFECT FOUND AND REPAIRED, C = DEFECT FOUND NOT REPAIRED, D = N/A<br>⚠ = TAKES MACHINE OUT OF SERVICE 💧 = CONDITION MONITORING POINT | | A | B | C | D | ⚠ 💧 |
|---|---|---|---|---|---|---|
| 173 | Remove lock pin and lower cabin, ensure it is secured | | | | | |
| 174 | Remove padlock & danger tags & re-energize equipment | | | | | |
| 175 | Remove articulation lock, box up lock, lower equipement to the ground and remove barricade | | | | | |

## START UP

| | | | | | | |
|---|---|---|---|---|---|---|
| 176 | Start unit and inspect for any fluid leaks | | | | | |
| 177 | Verify all fluids levels | | | | | |
| 178 | Notify operations when work is completed | | | | | |

## SIGN OFF

**TECHNICIAN NAMES:**

**TECHNICIAN SIGNATURES:**

**SUPERVISORS NAME:**

**SUPERVISORS SIGNATURE:**

**DATE:**

**DEFFICIENCY LIST**
**CATERPILLAR SCOOP (LHD)**
**HR - PM**

| Step No. | Defficiency | Corrective Action Required | Work Request No. | Comp |
|---|---|---|---|---|

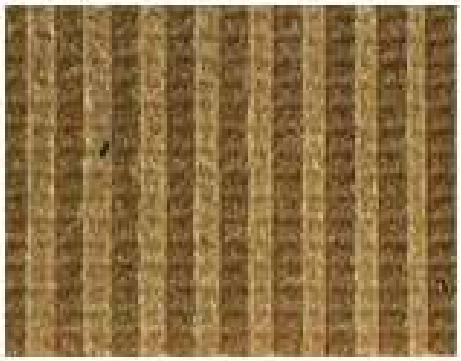
# PM FEEDBACK FORM

| HAVE YOU FOUND ANY PARTS DISCREPANCIES? | | | |
|---|---|---|---|
| ***STOCK #*** | ***INCORRECT PART #*** | ***CORRECT PART #*** | ***DESCRIPTION OF PART*** |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

| HAVE YOU IDENTIFIED ANY MAINTENANCE TASK TO BE UPDATED, ADDITION OR DELETED? | |
|---|---|
| ***TASK DESCRIPTION:*** | ***UPDATE / ADD / DELETE (circle one)*** |
| | |
| | |
| |

| | | |
|---|---|---|
| | **RATING 1** | **APPEARANCE:**<br>Small amount of fine grey paste. No shiny chips of metal or metal fragments.<br>**ACTION:**<br>Return to service after plug has been cleaned. |
| | **RATING 2** | **APPEARANCE:**<br>Small shiny metal particles. No metal chips or large fragments.<br>Advanced stages of fines showing early stages of bearing race and roller wear.<br>**ACTION:**<br>Return to service after plug has been cleaned.<br>Record Rating 2 on service sheet for closer monitoring over coming period. |
| | **RATING 3** | **APPEARANCE:**<br>Small shiny metal particles/chips.<br>More advanced stages of bearing race and roller breakdown. If final drive, check differentials magnetic plug and vice versa to ensure the other component is not the source.<br>**ACTION:**<br>Return to service after plug has been cleaned.<br>Notify maintenance supervisor. Close monitoring to continue. |
| | **RATING 4** | **APPEARANCE:**<br>Heavy covering of metal fragments with small to large chips.<br>Well-advanced failure of components if this amount of contaminants has accumulated since last inspection. If final drive, check differentials magnetic plug & vice versa to ensure the other component is not the source.<br>**ACTION:**<br>Notify maintenance supervisor immediately.<br>Inspection of failing component to be carried out ASAP.<br>Decision to return machine to service is at supervisor's discretion. |
| | **RATING 5** | **APPEARANCE:**<br>Very heavy covering of large shiny chips and slivers.<br>Failure has occurred if this amount of contaminants has accumulated since last inspection and is not residue from a previous failure. If final drive, check differentials magnetic plug and vice versa to ensure the other component is not the source.<br>**ACTION:**<br>Notify maintenance supervisor immediately so that repairs can be organised. **Machine must not be returned to service.** |

FILTER CONTAMINATION SCALE



1 No debris

2 Slight debris

3 Heavy debris

4 Extreme debris